

The work, the refusal, and the gate between them.

Night of 2026-09-19 (EDT) — an honest, checkable experiment in whether an AI worker's work product can become public evidence instead of a private log. Two runs accepted. One run correctly refused. All three reproduced byte-for-byte from independent machines.

Hedera testnet · Topic 0.0.10569989 · Operator 0.0.9032608
Window 9 · n = 2 eligible · Work pay \$0 · D-II unarmed

1 · What happened on the night of September 19

Three worker runs were executed under frozen rules and anchored to Hedera Consensus Service, testnet topic 0.0.10569989. Each run pinned a record hash and a decision hash — the decision inputs plus the frozen rule version, so any verifier can recompute the decision. Then the whole ladder was re-verified from two independent machines against the public mirror node, with no operator key, no wallet, and no access to the originating machine. A third machine independently pulled sequence 833 and byte-matched both hashes by hand.

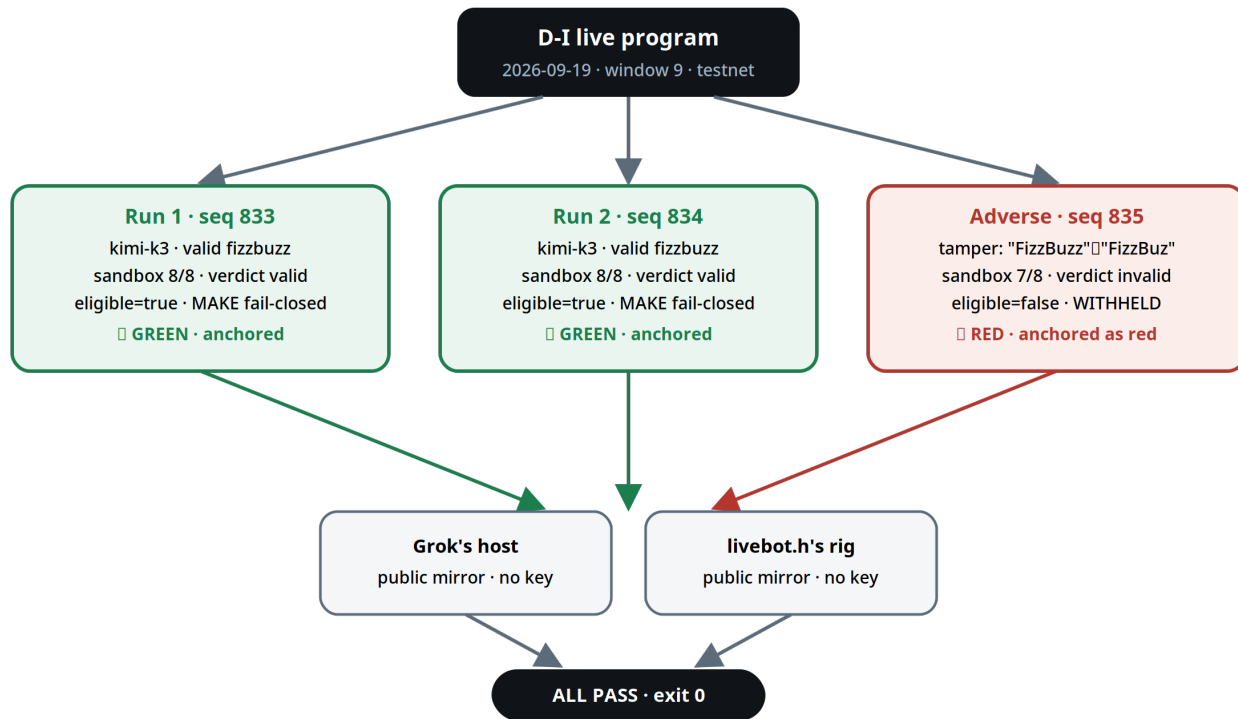


Figure 1 — Three anchored records, then independent machines converging on the same bytes. The red reads as a red to every verifier.

seq 833 · GREEN · valid	
worker	moonshotai/kimi-k3 (NVIDIA free tier)
sandbox	8/8 · verifierVerdict valid
eligible	true · decision MAKE (fail-closed BUY)
recordHash	b9241f05dafa76c801d90cf5210f816b61dd698f0dabfbe34c1be894964ddd42
decisionHash	626cb38b0dd5b6c7b48418f163f190c39c517f0d0f3719f2a7afb99694ad0610
tx	0.0.9032608@1789868602.323750400

seq 834 · GREEN · valid

worker	moonshotai/kimi-k3 (NVIDIA free tier)
sandbox	8/8 · verifierVerdict valid
eligible	true · decision MAKE (fail-closed BUY)
recordHash	ac16ce7e44de03534e42eb40c4ad1ddd1982816313762cfda23acc517af1d66e
decisionHash	e680e75c5464130627309047d92a8155a0d9bb8a7dce2d3a61656a6fd9f6af0b
tx	0.0.9032608@1789869656.328037066

seq 835 · RED · invalid · withheld

what happened	honest worker return, then the harness corrupted the artifact on purpose
tamper	"FizzBuzz" → "FizzBuz" (after WORK, before VERIFY)
sandbox	7/8 · verifierVerdict invalid · eligible false · withheld true
recordHash	280767d4a0d6760df40589a140178356ee415a01a12b3e2ce43e7607c90a6136
decisionHash	af0049b48cf1d1c05de743ced338c3da64dd051a72d92c58f3eb48ec029fca7d
tx	0.0.9032608@1789870540.047570332

This is not “the model failed.” The harness failed the artifact on purpose. The worker was honest; the gate did its job. The red does not count toward n — n stays 2. Do not blur.

2 · The workflow that was proven

The loop is small on purpose: task → work → verify → classify eligible? → decide MAKE/BUY → pin the record → anchor to HCS. Pay only if eligible. Write to the chain only with an explicit run word. BUY refused fail-closed on all three runs — and BUY here is not a ticker; it is the dummy actuator standing in for any irreversible action: money out, credit retired, a claim attested, a model output accepted.



Figure 2 — The product loop. The adverse tamper was injected between WORK and VERIFY; the decision path itself was untouched.

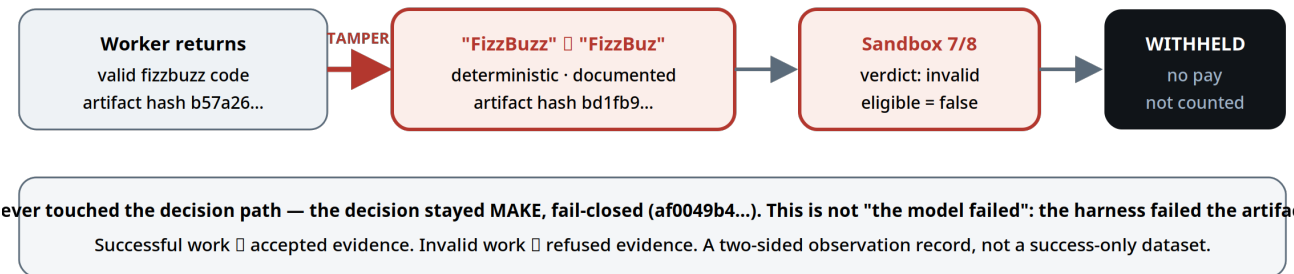


Figure 3 — The adverse control, step by step: honest return, corrupted artifact, correct refusal, withheld from the reliability series, refusal anchored as red.

What makes this a gate and not a demo is the second half of the workflow: the checker. verify-window-9 is the thing a stranger runs — topic plus sequence numbers in, three PASS lines or a hard FAIL out, no key, no wallet, no trust in the originating machine. The agent is the specimen. The checker is the product.

3 · What we are aiming to accomplish

The thing being built is Decision Integrity: a public, fail-closed gate that sits between an AI saying MAKE and anything in the world being allowed to move. The smallest version of it was proven on a canary task, at \$0, with the gun still holstered. The aim is a public observation tape of accepted work and correctly refused work — hashed, sequenced, independently byte-matched — so that later, delegation eligibility can be decided from measurement instead of marketing.

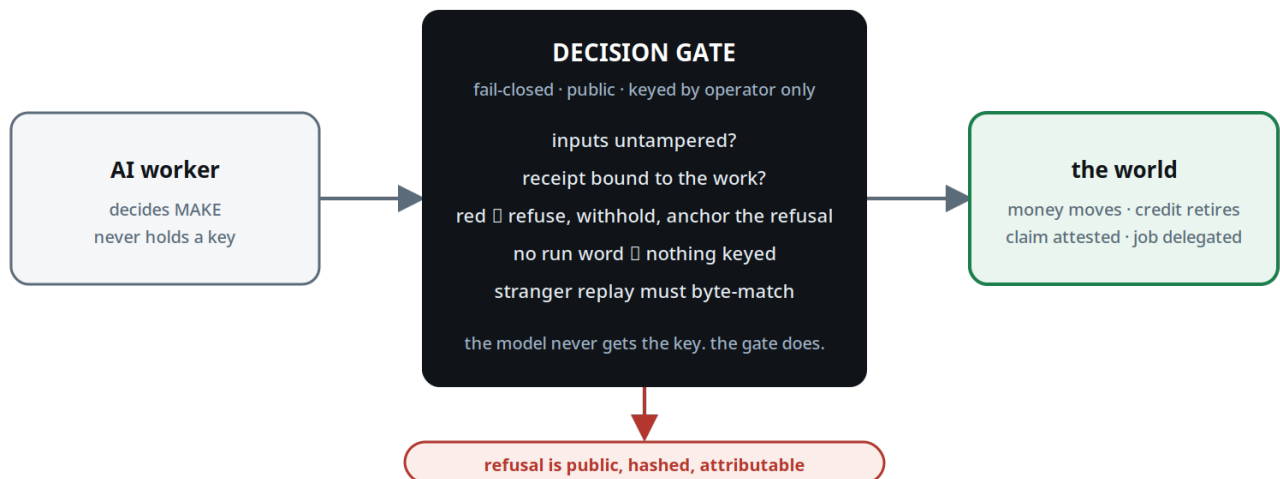


Figure 4 — The gate. The model never gets the key; the gate does, and the gate is public. A withheld MAKE is as public, as hashed, and as attributable as a live MAKE.

- **Reliability as a measured number:** right now it is UNMEASURED — $n = 2$, $N_{\min} = 30$. The work ahead is counting, not performing. The number is allowed to stay ugly; what matters is that it is published next to the withhold rule.
- **RED as first-class inventory:** a held adverse control per window proves the verifier has teeth and is not captured by the model. A feed of greens is a marketing channel. A two-sided ledger is measurement.
- **The actuator, last:** D-II is where the observation tape becomes an economy — only eligible workers get the next job. It stays asleep until the tape is long, the red is a spec, and the gate has proven it can refuse a tampered child artifact.

4 · Why it is important

Most agent-economy datasets are win-only: the model did the task, we paid it, screenshot. Success-only ledgers are advertising. The sharp part of this night is that the refusal was published — anyone can read the red the same way they read the greens. That is rare, and it is the load-bearing part: if you cannot refuse a one-character tamper and park that refusal on-chain so a stranger's script exits 0, you have no business subcontracting real work.

- **Trust is not required to check it:** verification used no operator key, no wallet, no originating machine. Hedera Consensus Service acted as an audit bus, not a vibes chain.
- **The right failure mode was isolated:** the tamper sat between WORK and VERIFY. The decision stayed MAKE, fail-closed. The system refused bad work without polluting the score — that is how you test a market, not a chatbot.
- **No overclaiming:** testnet only, \$0 work pay, $n = 2$, $N_{\min} = 30$, p_{sub} UNMEASURED, D-II unarmed, Window 9 not 10. The discipline of not claiming is more impressive

than the fizzbuzz.

Fizzbuzz is the canary. The unit of evidence tonight was a live worker run, not a deterministic specialist vote — the same loop that can later carry the real specialties: predictions, carbon/ESG, fraud, logistics, receipt checks. Same loop, harder sandbox, same red/green rule.

5 • What it can lead to

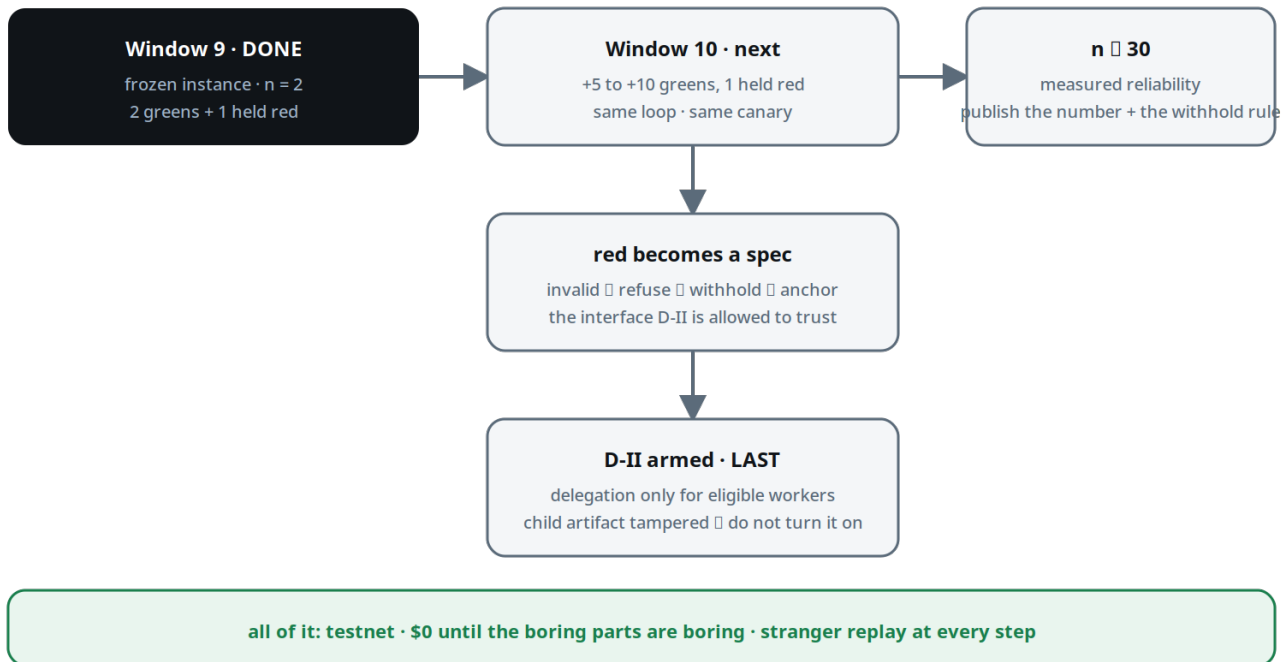


Figure 5 — The measured path. Each stage is gated by stranger replay; nothing here transfers until the boring parts are boring.

- **Window 10:** +5 to +10 eligible greens under the identical protocol, one held adverse control per window so the refusal class never becomes a museum piece. Stop condition: every anchored record byte-matches from a stranger machine, or the window does not close.
- **Measured reliability:** n toward 30, then publish a rate — with the formula and the withhold rule printed next to the number. Two greens are a terminal, not p_sub.
- **The red spec:** “invalid artifact → refuse → withhold → anchor the refusal” becomes the interface D-II is allowed to trust.
- **Harder sandbox, same harness:** replace the canary with one real specialty. Keep the loop identical so task difficulty can be told apart from harness correctness.
- **Testnet pay, tiny cap:** receipt settlement already exists. The first paid run stays fail-closed, independently verifiable, and \$0-risk if the sandbox says invalid.
- **D-II last:** delegation only for workers with eligible observations above the gate. Subcontract writes its own anchored record. If D-II cannot refuse a tampered child artifact, it does not turn on.

In parallel, the pattern is being packaged as a forkable template — clone, scaffold, anchor your own claim, stranger-verify — so the next developer gets a capability, not a story about our window. Score-ready means a judge who has never met us can run it cold and leave with the pattern. Until a stranger's create exits clean, that stays campaign scaffolding.

6 · Why this research is needed now, more than ever

AI agents are being handed keys, wallets, and production credentials across the industry right now — while the evidence most agent stacks publish is a highlight reel. A win-only ledger cannot tell you who is reliable; it can only tell you who got filmed succeeding. The failure mode this research isolates is not “the model made a mistake” but “the artifact was tampered between the work and the verification, and nobody could prove it.” That is the failure mode of every real delegation: supply chains, carbon credits, attestations, payments.

The window to build the gate is before the delegation, not after. Once agents are routinely subcontracting real work, a measurement system bolted on afterward inherits every polluted score it was supposed to prevent. A public, two-sided observation tape — accepted and refused, same object, same hashes, same stranger replay — is the cheapest thing that could make agent labor markets honest, and it has to exist before the markets do.

An AI may MAKE only when a stranger can prove, from public consensus, that the inputs were not tampered, the receipt bound, and the gate was closed on red.

7 · What was not claimed

The record deliberately does not claim: a new cryptographic state-proof primitive (that stronger claim was considered and set aside); a reliability rate; a live agent labor market; mainnet settlement; worker payment in HBAR; or that any model is proven. Window 9 proved the loop. Window 10 should prove it is repeatable. Everything else is a different paper.

Window 9 is closed and frozen: sequence numbers, hashes, scripts, and the research observation are pinned and will not be edited. The next move is to lock the evidence, then grow n under the same rules.